NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

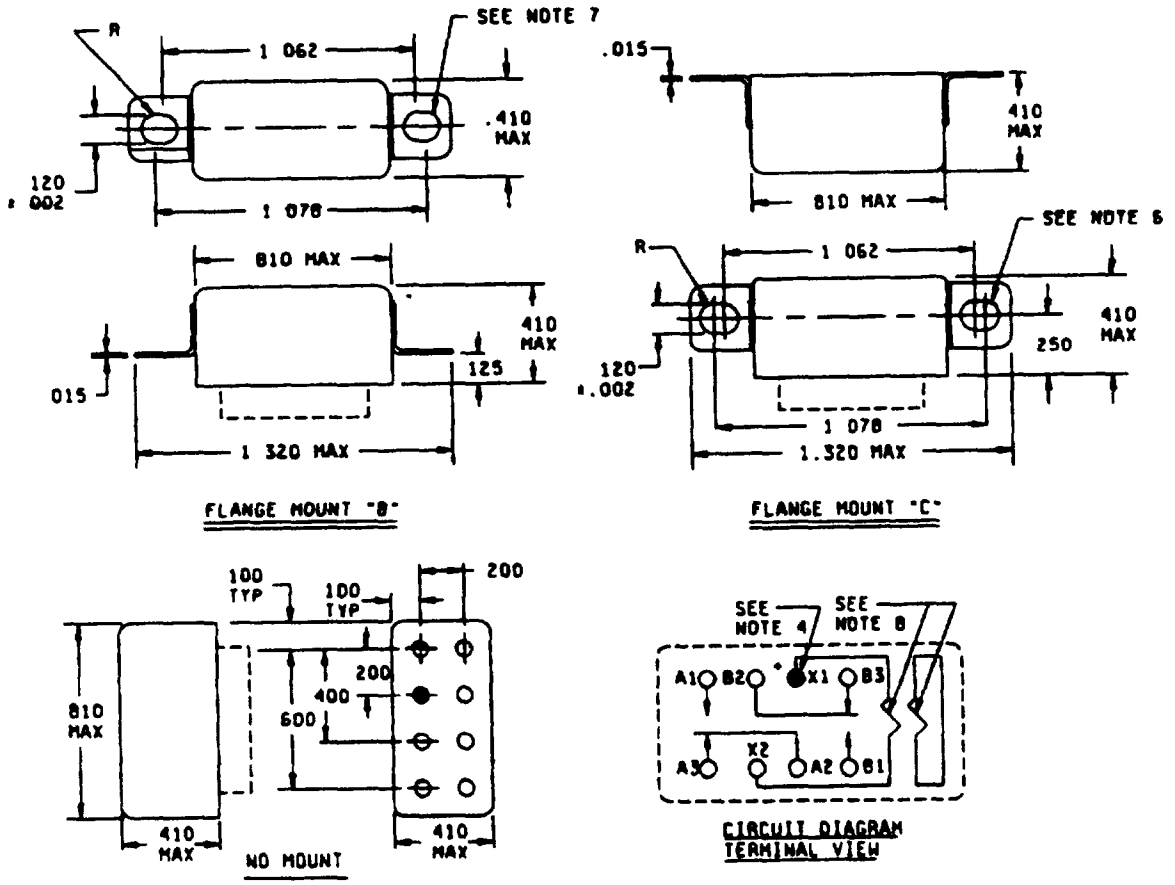
MIL-PRF-39016/22D
20 JULY 1988
SUPERSEDING
MIL-R-39016/22C
19 June 1981

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2 AMPERES (0 200-INCH TERMINAL SPACING) COIL TRANSIENT SUPPRESSED

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-R-39016



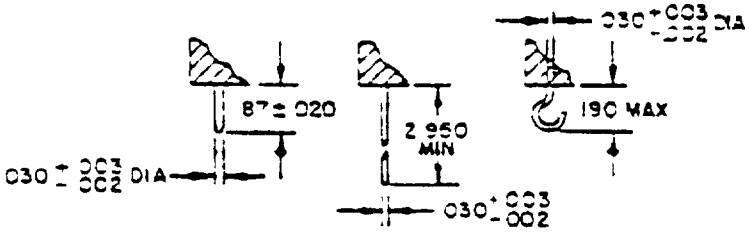
(D) FIGURE 1 Dimensions and configuration

(D) denotes changes

AMSC N/A 1 of 7 FSC 5945

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.020	0.51
.030	0.76
.100	2.54
.120	3.05
.125	3.18
.187	4.75
.190	4.83
.200	5.08
.250	6.35
.400	10.16
.410	10.41
.600	15.24
.810	20.57
1.062	26.97
1.078	27.38
1.320	33.53
2.950	74.93

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Indicated terminal shall be marked with a contrasting bead.
5. Terminal numbers A1 and A3 shall be marked on the relay.
6. Circuit diagram marked on top if legible from the mounted position, otherwise marking surface is optional.
7. Mounting screw head clearances based on use on no. 4 fillister head screws.
8. Coil symbols are optional per MIL-STD-1285.

(D) FIGURE 1. Dimensions and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

CONTACT DATA

Load ratings:

High level (relay case grounded)

Resistive: 2 amperes at 28 V dc, 100 ampere maximum at 115 V ac, (60 and 400 Hz). For qualification testing, 400 Hz loads not required.

Inductive: 50 ampere maximum at 28 V dc maximum with 200 millihenries minimum.

(D) Lamp: 160 ampere maximum at 28 V dc maximum

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac

Intermediate current: Applicable

Contact resistance and voltage drop

(D) Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: .100 ohm maximum.

Low level:

During life: 33 ohms maximum

(D) After life: 0.150 ohm maximum.

Intermediate current:

During intermediate current: 1 ohm maximum.

(D) After intermediate current: .300 ohm maximum.

Contact bounce: 2.0 milliseconds (ms) maximum. (Applicable to failure rate level "L").

Contact stabilization time: 2.5 milliseconds (ms) maximum. (Applicable to failure rate level "M", "P", and "R").

Overload (high level only): Two times rated current.

(D) COIL DATA. See table I.

Operate time: 5 ms maximum over temperature range with rated coil voltage.

Release time: 5 ms maximum over temperature range from rated coil voltage.

ELECTRICAL DATA.

(D) Insulation resistance: 10,000 megohms minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage

	Sea level Vrms (60 Hz)	Altitude Vrms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	1,000	350 All terminals to case
Between case, frame, or enclosure and coil(s)	500	
Between all contacts and coil(s)	1,000	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	1,000	

(D) COIL TRANSIENT CHARACTERISTICS. 1/

Maximum negative transient See table I.

ENVIRONMENTAL DATA:

Temperature range -65° to +125°C

(D) Vibration (sinusodial). MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

(D) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only.

(D) Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and, 1 microsecond maximum closure for open contacts

Magnetic interference Applicable.

Resistance to soldering heat Applicable.

Acceleration Applicable.

PHYSICAL

Terminals See figure 1 and table I.

Terminal strength: 3 ±0.3 pounds pull

(D) Solderability: Applicable.

Terminal twist test. Applicable to wire leads.

Dimensions and configuration: See figure 1.

Weight: 9 92 grams (0.35 ounce) maximum.

Identification marking (full): Applicable

(D) 1/ The following test shall be added to all qualification and group A testing performed on relays supplied to this specification. In all tables, sequence after coil resistance. Coil transient suppression: Apply rated voltage to coil using a cathode ray oscilloscope to monitor the voltage across the relay coil, instantaneously remove the coil voltage (or current) with a mercury switch and observe the monitor to verify that any transient generated does not exceed the specified value. Repeat the test applying opposite polarity to the coil For qualification only, legible photographs are required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.

LIFE TEST REQUIREMENTS.

(D) High level: 100,000 cycles per relay

(D) Low level: 100,000 cycles, plus 900,000 cycles mechanical life.

PART NUMBER: M39016/22-(dash number from table I and suffix letter designating failure rate level)

(D) TABLE I. Dash number and applicable characteristics. 1/

Dash number 2/			Mount	Coil voltage V dc 3/		Coil resistance ohms ±10%	At 25°C				Over temperature range		
Solder lug	Wire lead (SP)	Wire lead		Rated	Max		Specified pick-up value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pick-up value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Max coil transient voltage
004	005	006	Flange "B"	26.5	32.0	700	15.0	8.0	1.5	20.0	14.0	1.0	48.0
007	---	008	Flange "C"										
---	009	010	No mount										
011	012	013	Flange "B"	12.0	15.0	160	7.2	4.0	0.7	9.6	5.8	0.50	24.0
014	---	015	Flange "C"										
---	016	017	No mount										
018	019	020	Flange "B"	6.0	7.5	40	3.6	2.0	0.35	4.8	2.9	0.25	12.0
021	---	022	Flange "C"										
---	023	024	No mount										

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations). L, 3.0, M, 1.0; P, 0.1; R, 0.01. Example, 004L - - - - - 024R.

3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.

Ⓓ TABLE II. Qualification inspection and sample size. 2/

Single submission	Group submission	
20 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable.	M39016/22-004	20 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable.
	M39016/22-023	2 units, qualification inspection table, group II.
	M39016/22-013	2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal

1/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection testing, the relay manufacturer shall preselect the sampling plan.

2/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/6 may be used in addition to MIL-R-39016/22 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

QUALIFICATION INSPECTION (reduced testing): See table III.

If the relays produced for MIL-R-39016/22 are similar in construction and design except for the coil to the relays produced for MIL-R-39016/6, then reduced testing for qualification of MIL-R-39016/22 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/6 relays.

Ⓓ TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage - Group II of qualification inspection table.
1 unsealed sample unit - Internal examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
(D) Air Force - 85

Review activities:
Army - SL, AR
Navy - AS, OS, SH
(D) Air Force - 99
DLA - ES

User activities
Navy - MC
(D) Air Force - 11, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5945-0757-161)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:40Z
Check the source to verify that this is the current version before use.